Questions for the 15 Oct supervisor meeting: Framework v2 explained

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Three decisions in Framework v2 need Madam's view at the meeting (moved to Friday 16 Oct; planned for Oct 15, 2026). Each section below explains one in plain words, with examples from the Hanwella flood scenario, gives the options, and says what we recommend and why. Read the invariants section first: the three questions all depend on it.

The fidelity invariants in plain words

An invariant is a fact that must stay the same however the explanation is reworded. Profiles may change the order, the length, words versus pictures, text versus audio. They may never drop or change these five facts.

The forecast (synthetic) says: flooding at the Hanwella gauge on the Kelani Ganga, on Thursday, a 0.29 chance ("Possible") of reaching the Minor Flood Level, so get your go-bag ready today.

Invariant	The question it answers for the resident	In the Hanwella forecast
Hazard	What is coming?	Flood
Place	Where?	Here (Hanwella)
Time window	When?	Thursday
Likelihood (uncertainty kept)	How sure is it?	"Possible"; 3 in 10 houses or 29% if the profile shows them
Action	What do I do?	Get your go-bag ready today

The same forecast, explained three ways. All three pass, because all five facts survive:

Profile	What the screen says
Just tell me what to do	"Flooding is possible here on Thursday. Get your go-bag ready today." (read aloud)
Show me why	Same line, plus a picture of 3 of 10 houses and "Why? The ground is already wet after heavy rain."

Profile	What the screen says
How sure, and what changed?	Same line, plus 3 of 10 houses, "29%" on tap, and "Yesterday: 1 in 10"

Three explanations that `checkInvariants()` stops before the resident sees them:

What went wrong	Example wording	Check result	What the resident sees instead
A shortened version drops the uncertainty	"Flooding here on Thursday. Get your go-bag ready."	Fail: likelihood *missing*	The baseline explanation (L1)
AI rewording makes it sound more certain	"Flooding is likely here on Thursday…"	Fail: likelihood *wrong* ("Likely" is not "Possible")	The profile's own text (L2)
AI rewording gets the day wrong	"…possible here tomorrow…" (said on Monday)	Fail: time window *wrong*	The profile's own text (L2)

Why this matters: an explanation that is easier to read but silently drops "possible" can make a 3 in 10 chance sound certain. The check makes that impossible by design, and it runs on every explanation, not once at design time.

Question 1: which contribution leads the paper?

The paper has room for one headline software-engineering contribution; the other becomes supporting. We recommend leading with the invariants.

Contribution A, run-time fidelity invariants. The section above. The system guarantees that the five facts survive every adapted explanation, including AI wording, and checks it each time before display.

Contribution B, requirement → capability dependencies. Some explanation needs cannot be met by changing the screen; the system must first hold the right data. Examples from our scenario:

The resident asks	A screen change alone is not enough because…	So the system must have
"What changed since yesterday?"	You cannot show 1 in 10 → 3 in 10 without yesterday's forecast	Stored previous forecasts and a comparison (`compareForecasts()`)
"Is it above the old ferry steps?"	No landmark level exists anywhere for Hanwella	Local reference levels per area (a placeholder in our prototype)
"When is the next update?"	The model output has no schedule	Forecast schedule metadata (`nextUpdateAt`)

How the two compare:

	A: invariants	B: capability dependencies
New compared with related work	Yes: NeuroAdaptX checks equivalence once, by hand, before deployment, without uncertainty	Partly: NeuroAdaptX and Madugalla already trace requirements to features; the data side is our addition
Shown working in the prototype	Yes: built 14 Oct; `checkInvariants()` passes all 10 test cases and will run on every forecast screen	Partly: two of seven capabilities are placeholders
Easy to explain in one sentence	Yes	Needs the table
Risk	Reviewers may ask whether five invariants are enough	Reviewers may call it ordinary requirements analysis

Options to ask about:

1. **Lead with A, B supporting** (recommended). Strongest novelty, and fully buildable by the 29 Oct freeze.
2. **Present A and B equally.** Fuller story, but each gets half the space in 8 pages.
3. **Lead with B, A supporting.** Matches the Research Approach v3.1 wording ("this is the SE contribution"), but B is the weaker claim after the related-work check.

Question 2: what does alert mode show?

Alert mode (Layer 3) is the screen when a flood warning is live and people are under stress. Guideline G124 and our invariants rule disagree here, and we need Madam to choose.

What G124 says in our catalogue: while an alert is live, every profile shows "hazard, action and a call button only"; "Why?" returns when the alert ends. That drops three of our five facts: place, time window and likelihood.

The two versions side by side:

	G124 as written	Framework v2 proposal
Screen during an alert	"Flood. Get your go-bag ready." + call button	"Flooding possible here on Thursday. Get your go-bag ready today." + call button
Facts kept	Hazard, action	All five
Length	5 words	10 words
Invariants check	Must be switched off in alert mode	Runs as on every other screen

The evidence in our catalogue points both ways:

- **For keeping it short:** Mathur et al. (2026, CHI) found older adults wanted "direct, task-focused and concise explanations" in emergency alerts; extra detail added load under imagined stress.
- **For keeping the likelihood:** Joslyn and LeClerc (2012) found advice alone did not improve decisions, but advice plus uncertainty did. Wood et al. (2018) found longer warnings cut information-seeking delay.

A concrete case: without the time window, a resident who gets "Flood. Get your go-bag ready." on Monday cannot tell whether to act tonight or on Thursday. Without "possible", a false alarm feels like a broken promise, which may weaken trust in the next warning.

Options to ask about:

1. **All five facts in one short line** (recommended). Keeps the rule with no exceptions; still short.
2. **Hazard, place and action only in alert mode.** Closer to G124; the check runs with a reduced list of three, and the paper says why.
3. **Follow G124 exactly** (hazard and action). Shortest, but the invariants rule then has an exception, which weakens Contribution A.

Question 3: build the AI wording layer, or describe it?

Layer 4 lets a language model reword the profile text, for example to make it warmer or simpler. It is optional, and its output is shown only if it passes the invariants check. The question is how much of it the prototype needs by the 29 Oct freeze.

Example of what Layer 4 would do with the "Show me why" text:

	Text	Check
L2 profile text	"Flooding is possible here on Thursday. Get your go-bag ready today."	Pass
Good AI rewording	"There may be flooding here on Thursday. Please get your go-bag ready today."	Fail as written: the CAP word "possible" is gone, so the check falls back to L2. This shows the check is strict
Bad AI rewording	"Big flood coming Thursday! Pack now!"	Fail: likelihood missing; falls back to L2

Options to ask about:

1. **Test the check on prepared rewordings, no live model** (recommended). We write 10–20 example rewordings, good and bad, run `checkInvariants()` on each, and report how many it catches. No API key, no cost, no network risk during the walkthrough, and it still shows Contribution A working on AI-style text.
2. **Build Layer 4 with a live language model.** Most convincing demo, but adds cost, setup time and an unpredictable part to the walkthrough, with 17 days to the freeze.
3. **Describe Layer 4 only, no test.** Least work, but then the paper's claim about AI wording rests on design alone.

The "good rewording" row is worth showing Madam: it is a fair sentence that still fails, because our check requires the CAP word. Ask whether that strictness is right, or whether the check should accept close synonyms such as "may".

How to ask in the meeting

Show the Framework v2 figure first, explain the invariants in one minute with the Hanwella example, then ask the three questions in this order. Write Madam's answer beside each box.

- [ ] **Q1.** "After the related-work check, we think the run-time invariants are our strongest contribution, because NeuroAdaptX only checks equivalence once, by hand, and without uncertainty. Should the paper lead with the invariants and keep capability dependencies as supporting, or present both equally?"
- [ ] **Q2.** "G124 says alert mode should show only the hazard and the action. That drops place, time and 'possible', which breaks our invariant rule. We propose one short line with all five facts. Is that acceptable, or should alert mode keep only hazard, place and action?"
- [ ] **Q3.** "For the AI wording layer, we plan to test the invariants check on 10 to 20 prepared rewordings instead of a live model. Is that enough for the paper? And should the check require the exact word 'possible', or accept close synonyms like 'may'?"

Bring: `Framework_v2.pdf`, this document, and Supun's one-slide XAI comparison with the agreement count from 12 Oct.